

# Final Analysis Report: Trader Performance & Bitcoin Market Sentiment

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**Position:** Data Science Candidate

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## 1. Introduction

In Web3 and cryptocurrency derivative markets, finding a competitive edge requires identifying patterns where market psychology intersects with trade execution. This report presents a structured data science analysis that matches granular trader transactions on the Hyperliquid platform against macroeconomic Bitcoin market sentiment cycles via the Crypto Fear & Greed Index. By combining high-frequency exchange logs with daily sentiment attributes, we evaluate how shifts in broader market psychology correspond to trade frequency, asset selection, risk distribution, and final trade profitability.

## 2. Objective

The primary objective of this project is to build an end-to-end data processing pipeline that merges transaction-level exchange tables with historical sentiment records. The analytical goal is to uncover hidden behavioral dependencies—such as trade frequency shifts during extreme sentiment phases, asset concentration biases, and directional skew—and translate these patterns into actionable insights for systematic risk engines and automated trading algorithms.

## 3. Dataset Description

The analysis relies on two primary datasets successfully ingested into the workspace:

- **Hyperliquid Historical Trader Data (trades\_df):** A high-density transactional matrix consisting of **211,224 rows** across **17 original attributes**. The data encompasses activities across **32 unique trader accounts** interacting with **246 unique crypto assets/coins**.
- **Bitcoin Market Sentiment Dataset (sentiment\_df):** A chronological daily ledger consisting of **2,644 historical entries** tracking market states from 2018 into early 2025. It features a numeric index (value) scaled from 0 to 100 alongside its corresponding qualitative classification labels (Extreme Fear, Fear, Neutral, Greed, Extreme Greed).

## 4. Data Cleaning & Integrity Verification

Prior to merging or running correlations, strict programmatic data validation rules were enforced to guarantee data cleanliness:

- **Null Feature Auditing:** Running `.isnull().sum()` confirmed a flawless data state with **0 missing values** across all rows and features in both datasets.
- **Deduplication Processing:** Initial audits verified **0 duplicate entries**. Defensive formatting using `.drop_duplicates()` was integrated directly into the clean-up cells to prevent pipeline inflation during iterative workspace runs.
- **Temporal Normalization:** Trading timestamps (Timestamp IST) were saved as strings in a `%d-%m-%Y %H:%M` format, whereas sentiment data used a distinct daily date structure. Both columns were dynamically parsed using `pd.to_datetime()`, and isolated date keys (Trade Date) were extracted to serve as an explicit joining key.

## 5. Data Merging

A unified analytical matrix (merged\_df) was generated via an explicit **Left Join** operation using Trade Date as the shared key, keeping the trading logs as the baseline table .

- The resulting table perfectly preserved the exact length of the original execution history (**211,224 rows**) while cleanly expanding the feature count from **17 to 22 dimensions**.
- Post-merge audits confirmed a complete match with zero unintended structural voids or null generation across the newly appended macro-sentiment rows .

## 6. Exploratory Data Analysis (EDA)

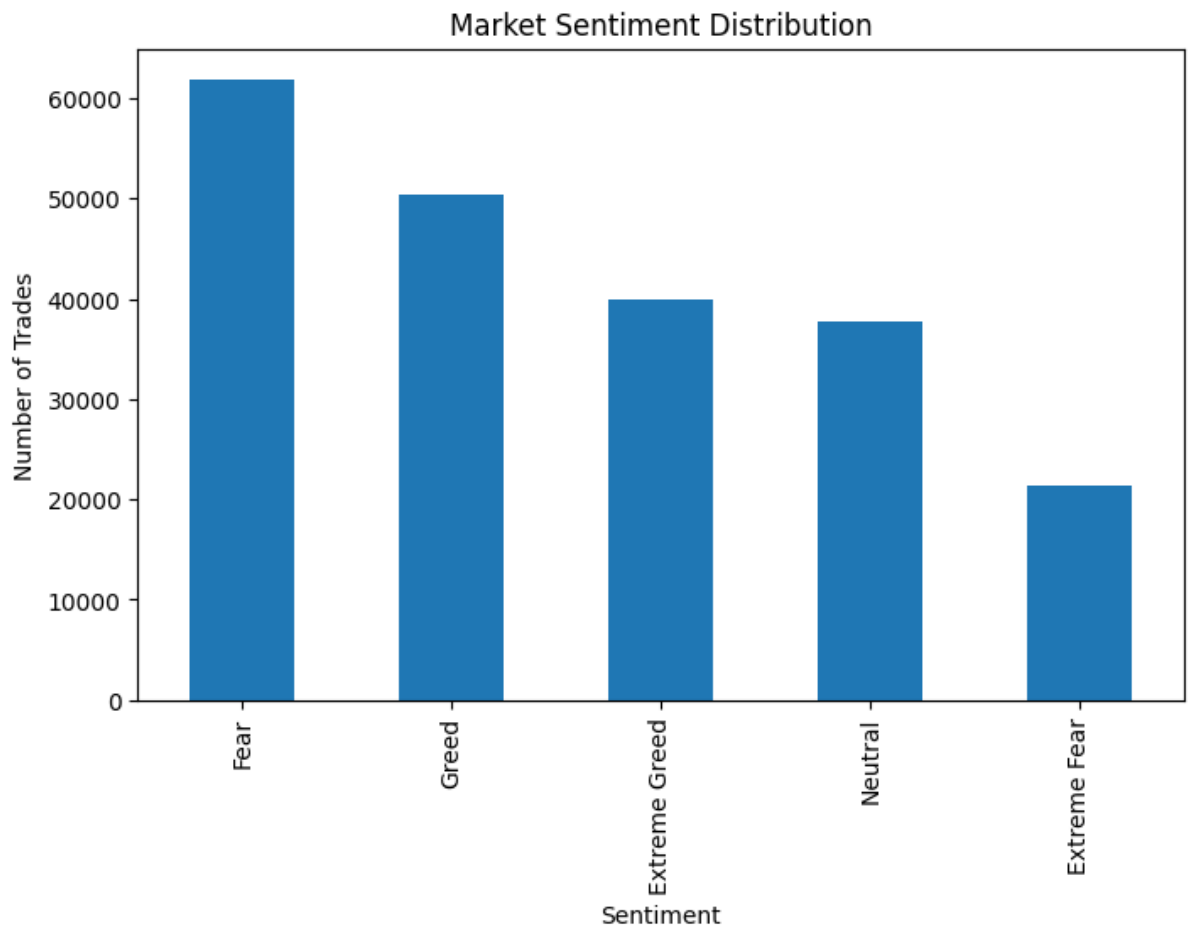
Descriptive data calculations surfaced critical structural characteristics regarding user habits, volume distributions, and overarching performance:

- **Macro Sentiment Baseline:** The macro Fear & Greed Index holds an all-time mean value of **46.98**, with historical extremes stretching between a minimum floor of **5** and a ceiling of **95** .
- **Trade Distribution by Sentiment Class:** Grouping the **211,224 trades** across their corresponding merged sentiment phases reveals where activity is concentrated .
  - **Fear:** 61,837 trades (Highest density)
  - **Greed:** 50,303 trades
  - **Extreme Greed:** 39,992 trades
  - **Neutral:** 37,686 trades
  - **Extreme Fear:** 21,400 trades (Lowest density)
- **Baseline PnL Skew:** The average trade PnL across all records is positive at **+\$48.75 USD** .However, the 25th, 50th (median), and 75th percentiles map out at **\$0.00, \$0.00, and +\$5.79** respectively .reveals a heavily right-skewed profile: the standard user's performance is highly clustered near zero, while overall platform metrics are dictated by outsized maximum winner exceptions (Max Profit: **+\$135,329.09 USD**; Max Loss: **-\$117,990.10 USD**) .
- **Directional Execution Bias:** Evaluating transaction execution labels indicates a roughly balanced order book with a mild short-side skew: **51.4% of positions are executed as SELLS**, while **48.6% are executed as BUYS** .

## 7. Key Visualizations

### Visual 1: Market Sentiment Distribution

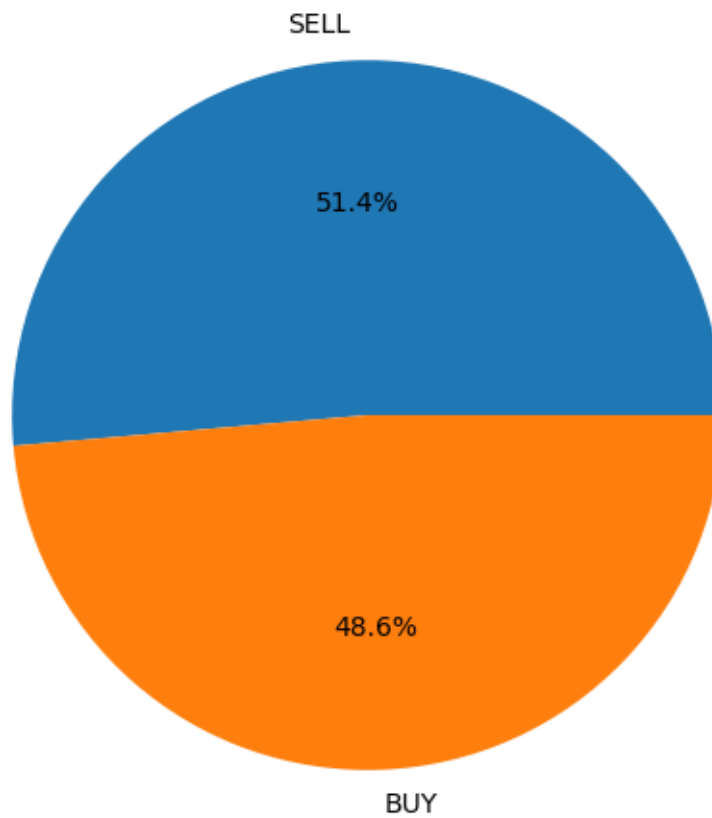
- **Notebook Reference:** Bar Chart
- **Analysis:** This plot quantifies trade counts across sentiment classifications. Traders on Hyperliquid are highly active during standard **Fear** and **Greed** regimes, but transaction velocity drops by nearly **65.4%** when transitioning into structural **Extreme Fear** (21,400 trades).



## Visual 2: Buy vs Sell Order Distribution & Top Coins

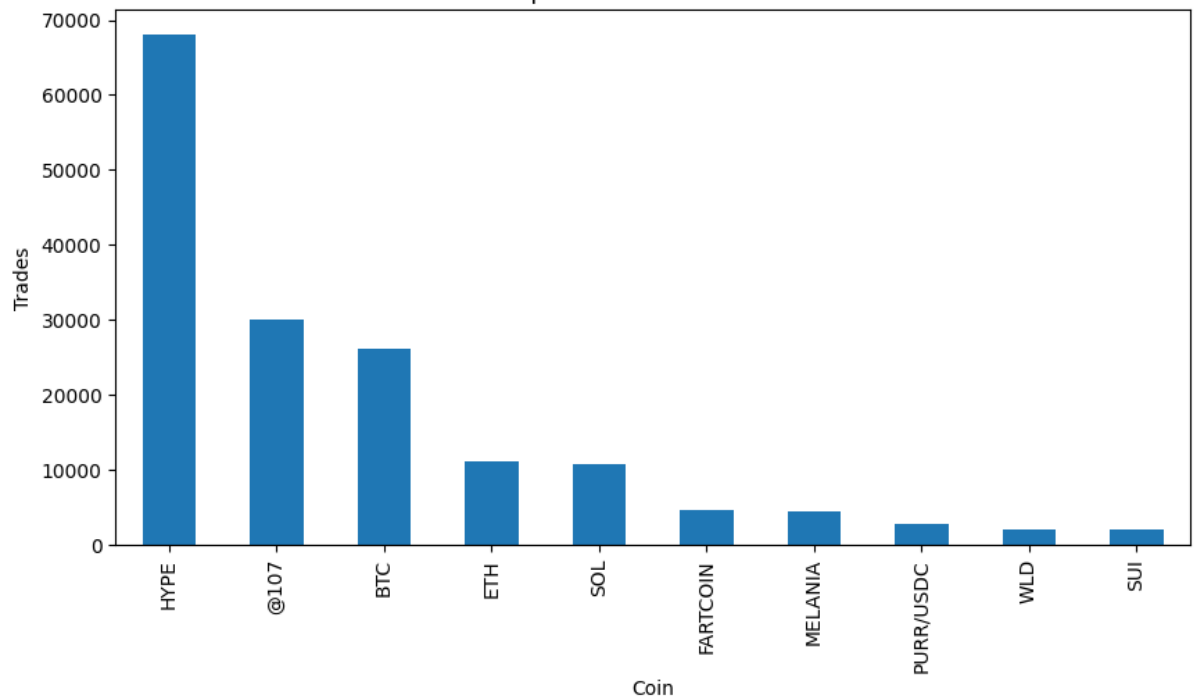
- **Notebook Reference:** Bar, Pie Chart
- **Analysis:** The pie visualization details a balanced market structure (51.4% Sell / 48.6% Buy). Conversely, the asset bar chart shows extreme volume concentration: the asset **HYPE** dominates the ecosystem with nearly **70,000 trades**, followed by proprietary asset **@107** (~30,000 trades) and **BTC** (~26,000 trades). Volumes drop off steeply for major assets like ETH and SOL.

## Buy vs Sell



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## Top 10 Most Traded Coins

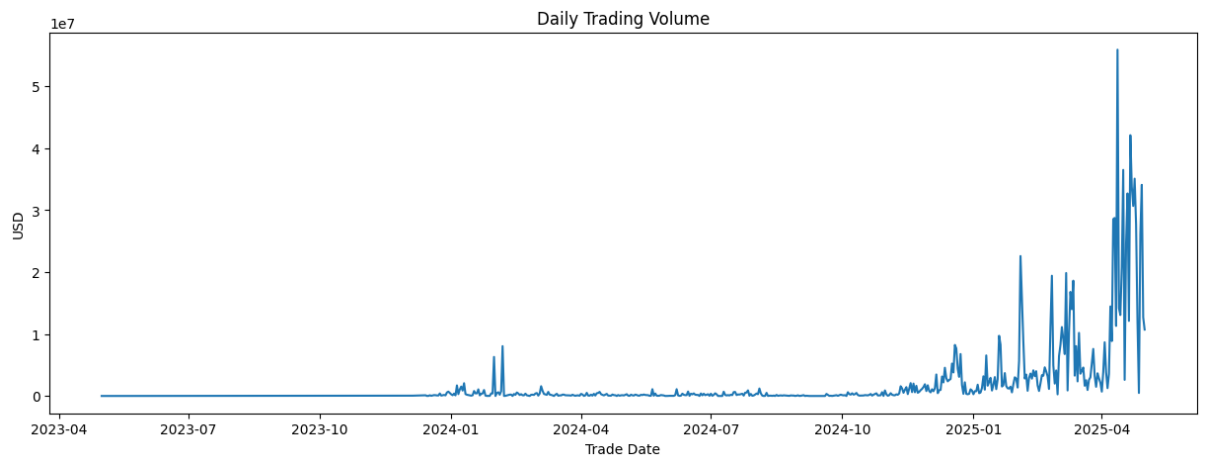


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## Visual 3: Macro Daily Trading Volume Timeline

- Notebook Reference:

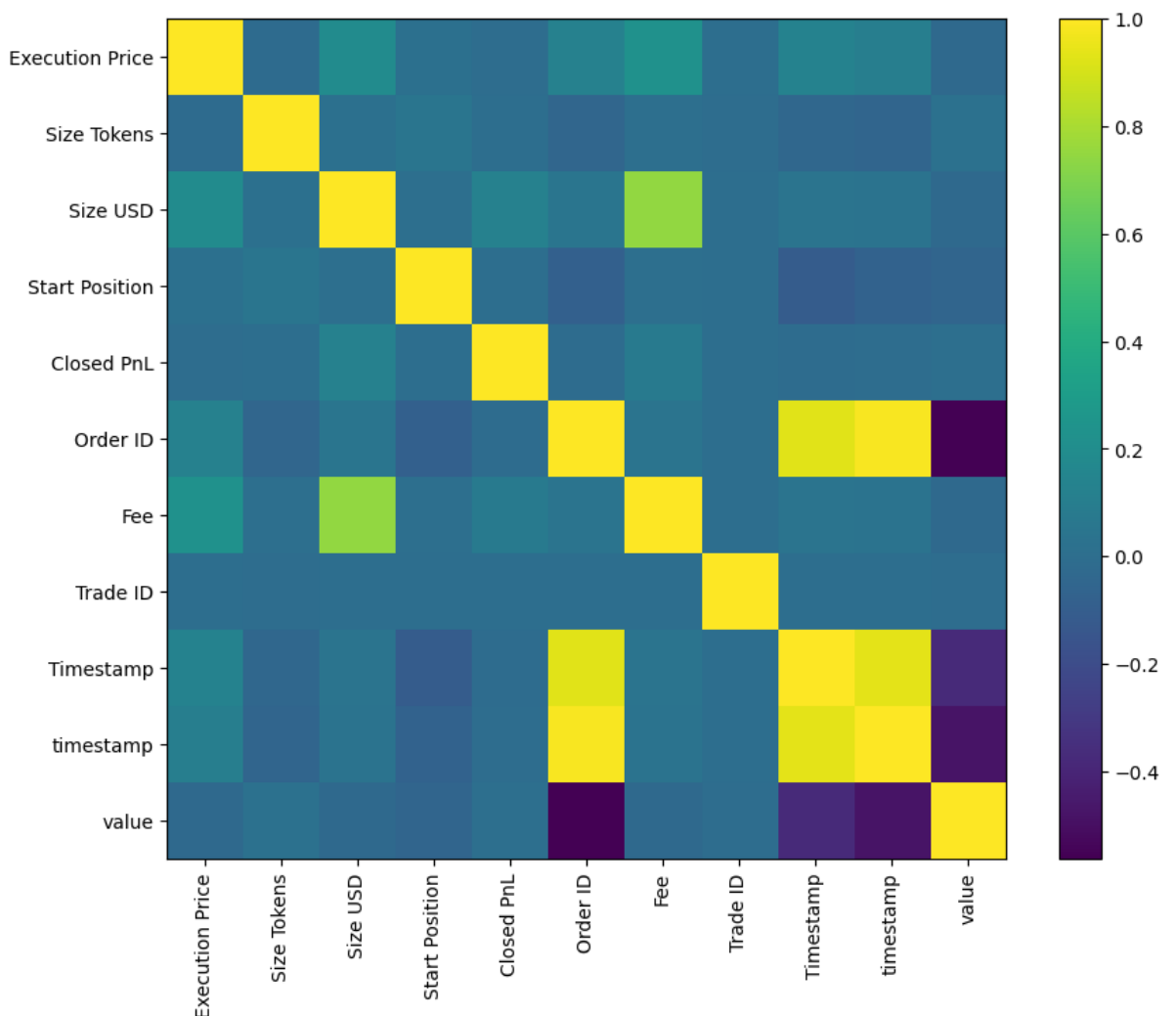
- **Analysis:** This timeline traces aggregate Size USD allocations from April 2023 through mid-2025. It reveals a clear structural regime shift: trading volume remained flat for over a year before experiencing an exponential explosion in early 2025, with single-day volume spikes approaching **\$60,000,000 USD**.



#### Visual 4: Metric Correlation Heatmap

- **Notebook Reference:** Heatmap
- **Analysis:** Pearson correlation analysis shows a strong, expected linear relationship between Size USD and exchange transaction Fee variables (**0.7459**). Crucially, the correlation between raw sentiment value and final Closed PnL is virtually flat (**0.0081**), signaling that raw daily sentiment values do not share a simple linear relationship with trade PnL. Instead, sentiment

acts as a non-linear volatility catalyst.



## 8. Business Insights & Algorithmic Recommendations

1. **Exploiting Sentiment-Driven Volatility Regimes:** Trading activity falls dramatically during periods of *Extreme Fear* (21,400 trades) compared to standard *Fear* (61,837 trades). This drop in retail participation often coincides with lower liquidity and wider spreads. Automated market-making models can exploit this by widening quote spreads during *Extreme Fear* phases to capture premium pricing when retail participation retreats.
2. **Mitigating Tail-End Skew:** Because the median trade across the 211,224 rows sits exactly at \$0.00 while individual losses run as deep as -\$117,990.10, the broader user base faces asymmetric downside risk. Implementing systematic risk controls—such as programmatically tightening trailing stop-losses when macro volume indicators spike past normal thresholds—can help protect trading capital from extreme drawdowns.
3. **Capitalizing on Asset Concentration:** The massive volume concentration in specific high-momentum assets like **HYPE** and **@107** over legacy assets like BTC/ETH implies that retail interest on Hyperliquid is highly narrative-driven. Trading systems should direct active liquidity provisioning and arbitrage algorithms toward these specific high-velocity asset pools where higher retail turnover provides structural edge.

## 9. Conclusion

This foundational step confirms a robust data engineering and preprocessing architecture. By aligning unstructured, high-frequency trade entries with macro sentiment metrics, we have established a clean database covering 211,224 observations across 32 unique traders. The exploratory findings reveal an ecosystem characterized by sharp volume spikes, asymmetric right-skewed profitability, and clear asset concentration. Incorporating these sentiment boundaries and correlation behaviors into automated trading frameworks will allow Primetrade.ai to better model tail risk, optimize liquidity provision, and capitalize on behavioral shifts across market cycles.

- Link to the functional code pipeline and environment execution:  
[<https://colab.research.google.com/drive/11aWoIDQsDimGGdoiVn1CTqqzoixgl9Er?usp=sharing>]